

Module 8: Cellular Genetics — Study Questions

1. What are the three stages of interphase? What happens in each?
2. What happens during S phase?
3. What is the difference between sister chromatids and homologous chromosomes?
4. List the four phases of mitosis in order.
5. What happens during metaphase?
6. What separates during anaphase of mitosis?
7. How many daughter cells does mitosis produce? Are they identical to the parent?
8. What is the difference between a cleavage furrow and a cell plate?
9. What is the purpose of meiosis? Where does it happen in humans?
10. How many daughter cells does meiosis produce? Are they diploid or haploid?
11. What separates during Anaphase I? What separates during Anaphase II?
12. What is crossing over? When does it happen?
13. What is independent assortment?
14. Why is it important that gametes are haploid?
15. Complete: Mitosis makes *cells that are*. Meiosis makes *cells that are*.
16. What is nondisjunction? Give one example of what it causes.
17. What are the three sources of genetic variation in sexual reproduction?
18. What are cell cycle checkpoints? Why are they important?
19. What is the difference between a proto-oncogene and an oncogene?

20. Why are siblings from the same parents not identical?